Part 1: Project Introduction

Project

Smart Library Management System

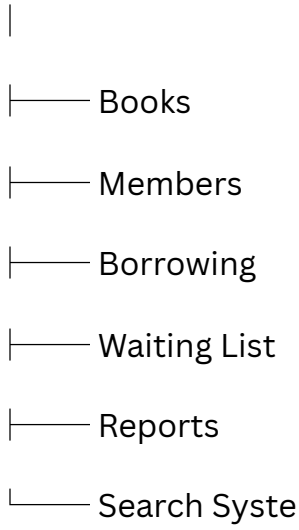
Goal

Build a complete application using everything learned so far:

- **Python**
- **OOP**
- **Searching**
- **Sorting**
- **Stacks**
- **Queues**
- **Linked Lists**
- **File Handling**

Part 2: Understand Requirements

Library



What information should a Book contain?

Book

{

book_id

title

author

year

copies

}

What information should a Member contain?

Member

{

member_id

name

phone

}

Borrow Record

BorrowRecord

{

member_id

book_id

issue_date

return_date

}

Part 3: Introducing OOP Design

```
class Book:
    pass
class Member:
    pass
class Library:
    pass
```

Why create classes ?

Book → handles all the operation of books

Member → Handels Members

Library → Controls Everything

- Real World mapping → OPP (Object Oriented Programing)

Part 4: Project Folder Setup

Install Required Tools

1. VS Code
2. Python 3.7+
3. Git
4. Other Libraires

Folder Structure

Create :

library_management/

```
│
├── main.py
├── models/
│   ├── book.py
│   ├── member.py
│
├── modules/
│   ├── search.py
│   ├── sorting.py
│   ├── borrow.py
│
├── dsa/
│   ├── stack.py
│   ├── queue.py
│   ├── linkedlist.py
│
├── data/
│   └── library.json
│
└── README.md
```

Part 5: System Flow Design

Start Program



What happens when user clicks Borrow Book ?



Yes → Issue Book

No → Add to waiting list (Queue)

Part 6: Database Design

For Now

Use JSON

Create:

```
{
  "books": [],
  "members": [],
  "borrowed": []
}
```

- JSON acts like a mini database.

Part 7: Assign mini Design Task

Create a class Diagram ..?

Book

Member

Library

Borrow Record

For each class:

Write:

- Attributes
- Methods

Example :

Book

Attributes:
---
book_id
title
author
year

Methods:
---
display()
update()

Part 8: Planning Further

Class 2

Book Management

Add Book
Delete Book
Update Book
View Books

Class 3

Member Management

Add Member
Delete Member
Search Member

Class 4

Searching Algorithms

- Linear Search
- Binary Search

Class 5

Sorting Algorithms

- Bubble Sort
- Selection Sort
- Insertion Sort

Class 6

- Borrowing System

Class 7

- Queue Implementation
- Waiting List

Class 8

- Stack Implementation
- Undo Feature

Class 9

- Linked List Implementation

Class 10

- Reports + Final Testing